

PLamb SNe Summary — KdS vs Λ CDM vs FR (V10V6c)

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Executive summary

KdS ($\Omega_\Lambda=0$, full N=1701): $\chi^2/\text{dof}=1.166684$, AIC=1987.86, BIC=2009.62, $w_{\text{eff}}=-0.780$ (median -0.810 -0.121/+0.123).

Λ CDM: $\chi^2/\text{dof}=1.166168$, AIC=1986.15, BIC=2002.47. FR ($\Omega_\Lambda=0$): $\chi^2/\text{dof}=1.166977$, AIC=1990.03, BIC=2022.66.

Evidence (lower is better): $\Delta\text{AIC}(\text{KdS}-\Lambda\text{CDM})=1.71$, $\Delta\text{BIC}(\text{KdS}-\Lambda\text{CDM})=7.15$; $\Delta\text{AIC}(\text{FR}-\Lambda\text{CDM})=3.87$, $\Delta\text{BIC}(\text{FR}-\Lambda\text{CDM})=20.19$; $\Delta\text{AIC}(\text{KdS}-\text{FR})=-2.16$, $\Delta\text{BIC}(\text{KdS}-\text{FR})=-13.04$.

KdS ($\Omega_\Lambda=0$) — parameters & metrics

Parameter	Value
H0	50.308662
Ω_m	None
Ω_Λ (fixed)	None
A_acc	1.037853
n_acc	0.659155
χ^2	1979.86
dof	1697
χ^2/dof	1.166684
AIC	1987.86
BIC	2009.62
N	1701
acc_frac	0.423000
w_{eff}	-0.780282
w_{eff} (median -/+)	-0.810 -0.121/+0.123

Λ CDM — parameters & metrics

Parameter	Value
H0	71.870108
Ω_m	None
Ω_Λ	None
χ^2	1980.15
dof	1698
χ^2/dof	1.166168
AIC	1986.15
BIC	2002.47
N	1701
acc_frac	0.646000

FR ($\Omega_\Lambda=0$) — parameters & metrics

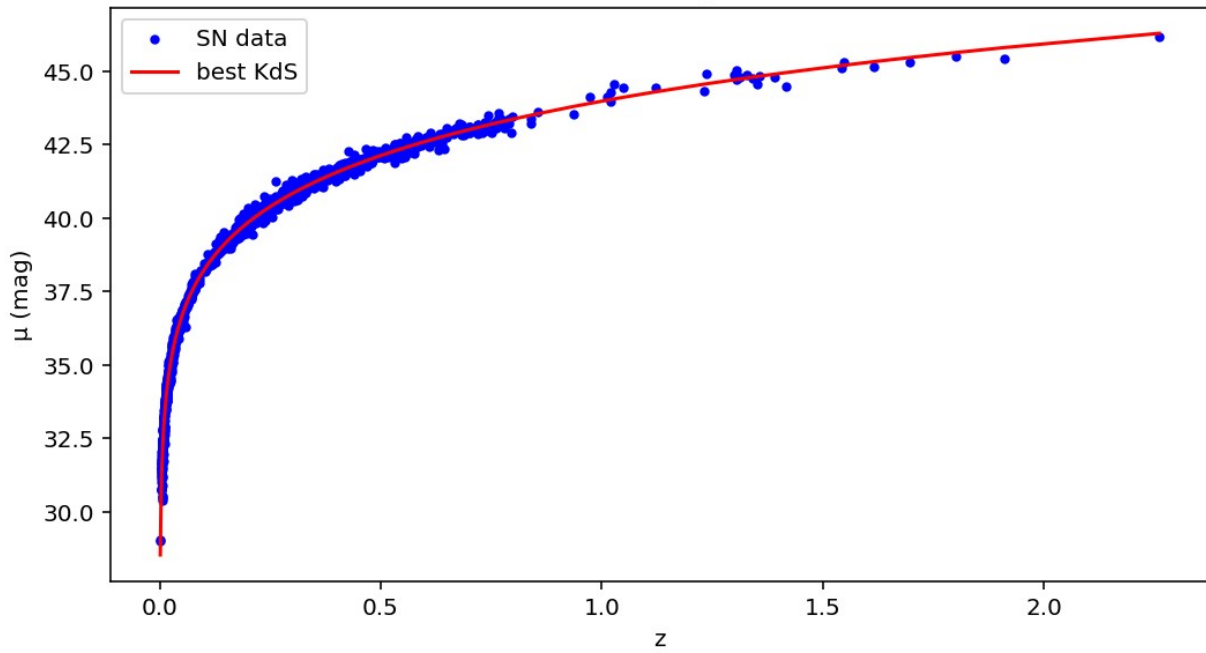
Parameter	Value
H0	60.646071
Ω_m	None
Ω_Λ (fixed)	None
A_acc	0.396099
n_acc	0.760047
χ^2	1978.03
Dof	1695
χ^2/dof	1.166977
AIC	1990.03
BIC	2022.66
N	1701

Figures (latest KdS run)

KdS: Hubble diagram

KdS | $H_0=50.31$, $\Omega_m=0.396$, $\Omega_\Lambda=0.000$, $A_{\text{acc}}=1.038$, $n_{\text{acc}}=0.659$, $\gamma_c=0.000$, $\epsilon_M=0.000$
 $\chi^2/\text{dof}=1.167$ ($k=4$, $\text{AIC}=1987.9$, $\text{BIC}=2009.6$) | frozen: $O_l=0$, $\gamma_c=0$, $\epsilon_M=0$

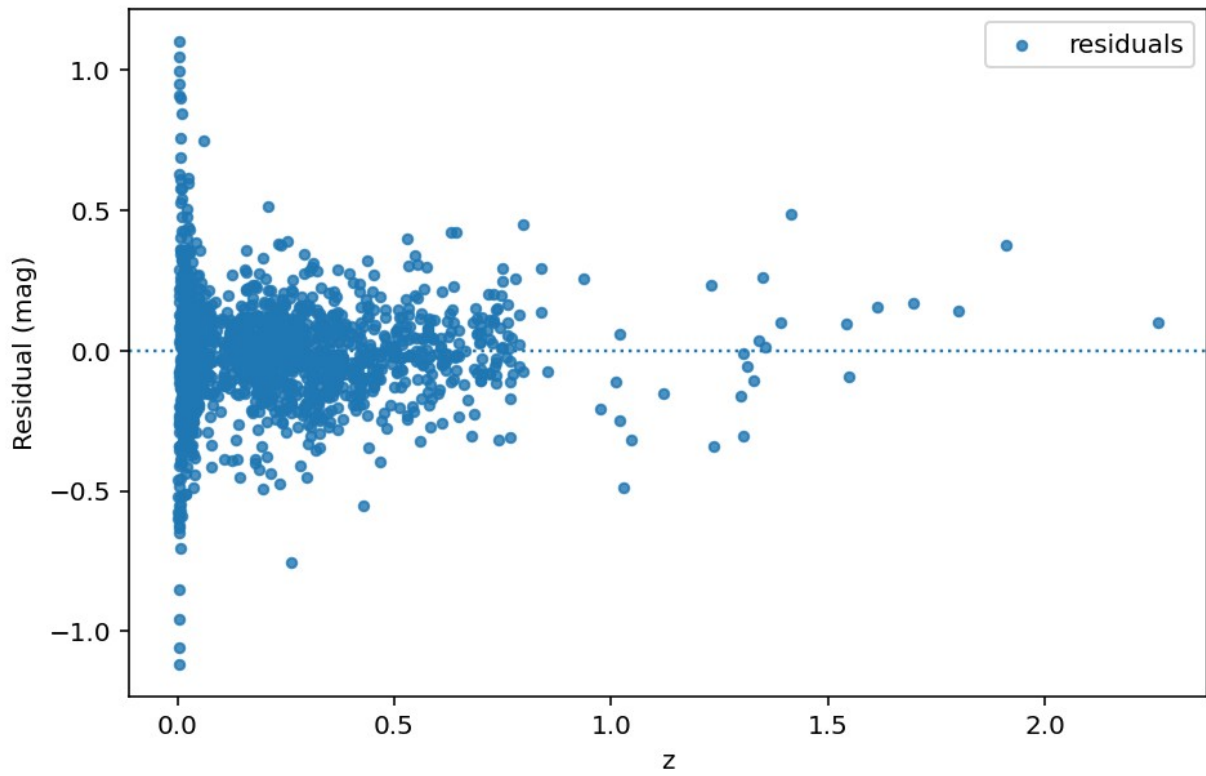
Hubble diagram



KdS: Residuals

KdS | $H_0=50.31$, $\Omega_m=0.396$, $\Omega_\Lambda=0.000$, $A_{\text{acc}}=1.038$, $n_{\text{acc}}=0.659$, $\gamma_c=0.000$, $\epsilon_M=0.000$
 $\chi^2/\text{dof}=1.167$ ($k=4$, $\text{AIC}=1987.9$, $\text{BIC}=2009.6$) | frozen: $O_l=0$, $\gamma_c=0$, $\epsilon_M=0$

Residuals



Interpretation

- KdS (no- Λ) achieves χ^2/dof comparable to ΛCDM on SNe (full covariance), but ΛCDM remains favored by AIC and especially BIC. KdS is favored over FR(no- Λ) by both AIC and BIC.
- The fitted KdS acceleration behaves as an effective fluid with $w_{\text{eff}} \approx -0.780$ (median $\approx \{\text{KdS}['w_{\text{eff_med']}]:.3f\}$), i.e., Λ -like but not exactly -1 .
- Posterior mean acceptance fractions were healthy (≈ 0.42 for KdS, ≈ 0.65 for ΛCDM).

PLambda Variable & Metric Glossary (V10V6c)

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Fit quality & model selection

Name (symbol)	Meaning	Units / domain	Notes
χ^2	Quadratic form of residuals using covariance: $(\mu_{th}-\mu_{obs})^T C^{-1} (\mu_{th}-\mu_{obs})$	mag^2 (dimensionless result)	Lower is better
dof	Degrees of freedom	$N - k$	N: #SNe used; k: #active params
χ^2/dof	Reduced chi-squared	≈ 1 ideal	Too high: underfit/underestimated errors; too low: overfit/overestimated errors
AIC	Akaike Information Criterion	$\chi^2 + 2k$	Lower is better; mild penalty on k
BIC	Bayesian Information Criterion	$\chi^2 + k \ln N$	Lower is better; stronger penalty on k
$\Delta\text{AIC}, \Delta\text{BIC}$	Differences vs a baseline model	dimensionless	0–2 weak, 2–6 positive, 6–10 strong, >10 very strong evidence against higher value
lnpost	Log-posterior at best sample	dimensionless	lnprior + lnlike; diagnostic only

Core cosmological parameters

Name (symbol)	Meaning	Units / domain	Notes
H0	Hubble constant today	$\text{km s}^{-1} \text{Mpc}^{-1}$	Sets distance scale
Ω_m (Om)	Matter density fraction today	0–1+	Dust-like, scales as $(1+z)^3$
Ω_Λ (Ol)	Cosmological constant fraction	0–1+	Often fixed 0 in no- Λ tests
Ω_k (Ok)	Curvature parameter	$-\infty \dots \infty$	Derived: $\Omega_k = 1 - \Omega_m -$

$$\Omega\Lambda$$

PLamb “extra physics”

Name (symbol)	Meaning	Units / domain	Notes
A_acc	Amplitude of effective acceleration component	dimensionless	Appears in $E^2(z)$ as $A_{\text{acc}} (1+z)^{\{n_{\text{acc}}\}}$
n_acc	Index of acceleration component	dimensionless	Implied fluid EoS: $w_{\text{eff}} = n_{\text{acc}}/3 - 1$
γ_c (gamma_c)	Variable-speed-of-light slope	dimensionless	$c(z) = c_0[1 + \gamma_c z]$; enters light-travel integral
ϵ_M (epsilon_M)	SN absolute-magnitude evolution amplitude	mag per z (linear_z) or per $\ln(1+z)$ (log1pz)	$\Delta M(z) = \epsilon_M \times f(z)$

Expansion & distances

Name (symbol)	Meaning	Units / domain	Notes
E(z)	Dimensionless Hubble rate	dimensionless	$E^2(z) = \Omega_m(1+z)^3 + \Omega\Lambda + \Omega_k(1+z)^2 + A_{\text{acc}}(1+z)^{\{n_{\text{acc}}\}}$
D_H	Hubble distance	Mpc	$D_H = c/H_0$; c may be replaced by c(z) in integrand
D_C	Line-of-sight comoving distance	Mpc	Integral of $c(z)/E(z)$
D_M	Transverse comoving distance	Mpc	Includes curvature via sin/sinh for $\Omega_k \neq 0$
D_L	Luminosity distance	Mpc	$D_L = (1+z) D_M$
μ (mu)	Distance modulus	mag	$\mu = 5 \log_{10}(D_L / 10 \text{ pc}) + \Delta M(z)$

Data & covariance

Name (symbol)	Meaning	Units / domain	Notes
N	Number of SNe used after masking/thinning	integer	Reported per run
C	Full SN covariance matrix	mag ²	Subset to mask; symmetrized; small jitter for SPD
C ⁻¹	Inverse (or pseudo-inverse) covariance	mag ⁻²	Used in χ^2
σ	Diagonal uncertainties	mag	Fallback if no full covariance
zmax	Maximum redshift cut	dimensionless	Mask: $z \leq z_{\text{max}}$
thin	Thinning factor	integer ≥ 1	Use thin=1 for apples-to-apples AIC/BIC

Sampler controls (emcee)

Name (symbol)	Meaning	Units / domain	Notes
sampler	Sampling mode	emcee grid none	Ensemble MCMC (emcee) recommended
nwalkers	Number of walkers	integer	$\geq 2-4 \times \text{\#active params}$
nsteps	Steps per walker	integer	Total chain length
nburn	Burn-in steps discarded	integer	Per walker
acceptance fraction	Mean accepted proposal fraction	$\sim 0.2-0.6$	Reported post-run
seed	Random seed	integer or None	Reproducibility
corner	Save posterior corner plot	flag	Also saves simple trace plots
show	Display figures inline	flag	Optional; saves by default

Numerical & bookkeeping

Name (symbol)	Meaning	Units / domain	Notes
nint	Base Simpson panels for integrals	integer	Grows with z for stability
dM-mode	Form of $\Delta M(z)$	linear_z log1pz	ε_M multiplies chosen $f(z)$
out	Output directory	path	Stores bestfit txt, plots, traces
active / frozen	Parameter status	lists	Affects k in dof/AIC/BIC
model	Chosen cosmology	LCDM KdS FR	LCDM freezes extras; KdS uses $A_{\text{acc}}, n_{\text{acc}}$ only; FR allows γ_c, ε_M

Quick interpretation cheats

- $\chi^2/\text{dof} \approx 1 \rightarrow$ statistically consistent residual scatter; $>>1$ underfit; $<<1$ may indicate overestimated errors.
- Lower AIC/BIC $\rightarrow$ preferred model; $\Delta\text{BIC} \geq 10 \rightarrow$ very strong preference for the lower-BIC model.
- $w_{\text{eff}} = n_{\text{acc}}/3 - 1$: $0 \rightarrow -1$ (Λ -like), $\approx -1/3$ (curvature-like), < -1 (phantom-like).
- Use thin=1 when comparing models via AIC/BIC; thinning changes N and hence the BIC penalty.
- Check acceptance fraction (~ 0.2 – 0.6) and inspect corner/trace plots for convergence and degeneracies.